Wenguang (Winston) Dong

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EDUCATION

Carnegie Mellon University, H. John Heinz III College Expected Dec 2026
Master of Information Systems Management (MISM, 16 months)

Hangzhou Dianzi University Sep 2021 – Jun 2025
Information Management and Information Systems

SKILLS

- Machine Learning: Transformers, RAG, PyTorch, Langchain, Dify, OpenCV, Multimodal, AI Red Teaming, AI Trend Analysis
- Programming/Database: Python, Pandas, Numpy, Java, SpringBoot, SQL, MySQL, PostgreSQL
- Deployment/Tools: Github, Docker, Postman, Charles, JMeter, APIFOX, CRM, FIgma, Axure, Firecrawl
- Data Analysis & Statistics: SPSS, Power BI, Statistical Analysis, Jupyter Notebook

PROFESSIONAL EXPERIENCE

AI Engineer Intern Dec 2025 – Present | Remote
ReelFluent Inc.
- **Built** an LLM-driven pipeline to extract, validate, and structure linguistic knowledge points from 100k+ SRT subtitles using Gemini API.
- **Developed** an end-to-end LLM-based pipeline cluster 40K+ knowledge points, generate master motives/specifics, and automatically package them into 70+ product-ready Lingo Packs with Gemini-generated metadata and SEO-ready branding.
- **Implemented** timestamp-accurate slicing, episode metadata generation, and FFmpeg-based screenshot extraction to support AI-powered language learning.
- **Developed** a new AI tutor module using Gemini APIs and exposed it as a Flask-based service for interactive chat.

Testing Engineer Intern June – Aug 2023 | Hangzhou, China
Xiaoqiang Parking, Hangzhou Jiahu Technology Co., Ltd.
- **Quality Testing:** Conducted buried point tracking tests (analyzed via Umeng), multi-concurrency stress testing (JMeter), and connectivity tests for client app and CMS APIs (APIFOX) to ensure system stability
- **Issue Identification:** Diagnosed and documented slow-loading pages (homepage, order page) and discovered login logic issues in the newly released test client app, providing timely feedback for fixes. Utilized Charles for network packet capture and Navicat for database verification
- **CRM/Business Integration:** Lead the internship team to successfully coordinate and test CRM system integration with 2 new parking lots, ensuring seamless operation and accurate data exchange
- **Teamwork:** Communicated with developers about the found test problems and assist in solving them; Participated in product requirement meetings and proposed a lottery-based feature design, which was adopted and integrated into the final product plan

ACADEMIC PROJECTS

Realtime Speech-to-Text Transcription App for macOS Nov – Present
https://github.com/DWGqaz123/realtime_transcriber_WG
- **End-to-End System Design:** Built a real-time speech transcription and intelligence desktop app with a FastAPI + native SwiftUI macOS, enabling low-latency streaming STT via ElevenLabs Scribe v2 Realtime API.
- **Audio Pipeline:** Engineered a realtime streaming pipeline using AVAudioEngine.
- **LLM-Powered Intelligence:** Designed and integrated an LLM-driven summarization stream that generates real-time knowledge cards, combining OpenAI-based summarization with RAG-based long-term memory using FAISS to support contextual recall, meeting summaries, and knowledge-aware transcription.
- **Scalable Architecture & Reliability:** Implemented double-buffer container architecture, asynchronous indexing, and semantic-triggered summarization to balance real-time responsiveness, data consistency, and output quality.

Dify-based Stardew Valley AI Encyclopedia (Chatbot) June – July 2025
- **Knowledge Engineering:** Built a curated Stardew Valley knowledge base from the official Chinese wiki; leveraged Firecrawl and custom crawler scripts to improve chunk quality, boosting retrieval accuracy by about 30% and reducing irrelevant hits by about 25%
- **Workflow Design:** Implemented a multi-step RAG pipeline, including intent recognition, query rewriting, retrieval and outcome evaluation, improving answer correctness and response completeness

RAG-based Intelligent Q&A Chatbot May – June 2025
Link: https://github.com/DWGqaz123/cmu_mism_rag_assistant
- **NLP & Knowledge Engineering:** Processed multi-format documents (HTML/PDF/TXT) with metadata tagging, cleaning, and semantic-preserving chunking; applied embeddings (HuggingFace model) to build a structured knowledge base using ChromaDB.
- **RAG Pipeline Design:** Engineered an end-to-end RAG workflow with query classification, hybrid retrieval (semantic vector search + BM25), and cross-encoder re-ranking, ensuring contextual accuracy and reducing hallucinations in answers. Applied testing, and evaluation.
- **Cloud & Model Integration:** Deployed pipeline across OpenAI API, Gemini API, and local Ollama models
- **Impact:** Created a reliable Q&A assistant with context-grounded, cited answers, improving retrieval precision and answer completeness for CMU MISM program documents

Language Learning Institution Management System Mar – May 2025
https://github.com/DWGqaz123/languagelearning_system
- **System Analysis:** Analyzed the operational pain points of small and medium-sized language training institutions in China, and designed a unified digital management platform to streamline course scheduling, assignments, mock exams, and resource sharing.
- **Frontend/Backend/Database Development:** Engineered a full-stack web application with 110+ RESTful APIs using Spring Boot, MyBatis-Plus, and React, enabling seamless interaction between frontend and backend. Architected and optimized a relational database schema in MySQL.